

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY
NTA LEVEL 5 - ORDINARY DIPLOMA IN ENGINEERING
CONTINUOUS ASSESSMENT TEST - I - JANUARY - 2026
EDT 05103 - APPLIED CHEMISTRY

SEMESTER: I

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

Auth: 4LAZIE

PART – A (Answer all the Questions)

(7 x 2 = 14)

1. State why most graduates in materials programs are trained to be both material scientist and materials engineers?
2. In addition to structure and properties what are the other two important components involved in science and engineering of materials?
3. Why alkaline dry cells last longer as compared to typical acidic cells
4. Define the term electrochemical series
5. What is overvoltage as applied in electrolysis
6. What is Faraday
7. What is Vulcanization process in rubber production

PART – B (Answer any three Questions)

(3 x 4 = 12)

8. Briefly explain the interrelationship between the structure and properties of materials
9. What are crystalline defects (Imperfection in solids), give at least three examples.
10. Mention the theoretical and practical importance of Electrochemistry
11. Differentiate Mesh number and Moh's scale in measurement of abrasives.
12. State why in practice you will preferably use Metal structures like screws, bolts, nuts, connected metal parts and rivets of the same metals, to make your system/product?
13. With examples write down three basic groups/categories of materials.

PART – C (Answer any two Questions)

(2 x 12 = 24)

14. i. What is periodicity as applied in atomic scale structure of materials.
ii. Describe the three primary categories of the atomic scale structure of materials based on their periodicity
15. (a) Name two industrial applications of electrolysis. (a) Calculate the time to deposit 1.56g of nickel using a current of 2.34A in a nickel-plating process. (Assume that $(Z) = \frac{1}{72000} \text{g/C}$ of nickel are deposited per ampere-seconds)
16. Explain at least six (6) measures/methods for corrosion control (protection against corrosion)
17. i. What are fuel cells?
ii. How do fuel cells differ from batteries?
iii. Describe a hydrogen fuel cell.